

# How do Archaeologists Identify Abundance?

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When archaeologists look for evidence of abundance, they focus on three elements: scale, rarity, and processing. As described by Gremillion, some of the most famous archaeological evidence of feasting involves the discovery of food remains (or culinary cookware or tableware) on a scale that exceeds that of a single household. For example, the masses of seeds, nuts, and bones in the sub-Mound 51 pit at Cahokia were deposited rapidly, in a large quantity of leftover trash. The masses of ceramics found under Mound 51, at the Lubbar Creek mound, and in the throne room of Pylos, Greece, extend well beyond what is needed for a household or even an extended family. They are the remnants of massive meals, feasts, that drew on food beyond that needed for everyday subsistence. Rare food items may be found in greater quantities in such contexts—for example, large fish and prairie chickens under Mound 51 at Cahokia.

Abundance can also be indicated by the use of processed food, rather than raw foodstuffs. The inhabitants of Moundville received maize kernels already separated from their cobs, indicating that the maize was grown and processed (dried, and the kernels stripped from the cobs) elsewhere. For the growers, that food was surplus. The inhabitants of Moundville, receiving this prepared food, were able to spend their time engaging in activities other than growing food. Surplus made those activities, and even specialization in specific trades and crafts, possible.

## Food Storage

One concrete archaeological sign of surplus is the scale of food storage. Archaeologists have excavated a wide variety of storage structures around the world that were used to hold durable foodstuffs and keep them under stable conditions for later use. Storage structures vary in technology and scale, but common features include restricted access (reducing the risk of pests, or even human thieves, accessing the food); weather-proofing; and locations that limit access to only those authorized to access the storage. Simple pits in the ground—especially if dug into bedrock (as in Maya *chultunes*) or rock-lined (as in Iron Age Israel), but even those dug into soil or sand (as in the Fayum basin of Egypt's Western Desert)—can preserve grains at least into the next harvest season, and sometimes for many years.



Types of storage pits. From top to bottom: Maya chultun (from La Blanca, Guatemala); rock-lined storage pit (from Negev, Israel, and now at Hecht Museum, Haifa); basket-lined storage pit (from Fayum, Egypt).

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In a provocative article, titled [“The Neolithic Refrigerator on a Friday Night: How Many People Are Coming to Dinner and Just what Should I Do with the Slimy Veggies in the Back of the Fridge?”](https://www.tandfonline.com/doi/full/10.1179/1749631415Y.0000000003) (<https://www.tandfonline.com/doi/full/10.1179/1749631415Y.0000000003>) Ian Kuijt makes the

argument that, when storing grain, farmers need to account for more than just the amount of time they expect between putting this year's harvest into storage and the beginning of the following year's harvest. Farmers need to plan for longer periods: *What if next year is a drought and I don't receive a harvest? I had better plan to store two years' worth of grain, or three if I can manage it.* They need to store grain needed not only for food but also to plant the next year's crop (and perhaps the crop for the following year). They need to account for losses in storage due to pests, mold, or other adverse conditions. Taking all three of these factors into account, a family of farmers needs to store, at a minimum, 1.4 times the amount they plan to need the following year. Adding in the potential for next year's crop to fail, it is necessary to plan for at least one year (better two, and even better three) of extra grain. If one year, the family needs to store a minimum of 2.8 times the amount they eat in a year. For two years of extra grain, they need to put away more than four (ideally 4.5) times the amount they eat in a year.

By this math, it becomes clear that large storage quantities were necessary even for farmers, so the amount available as surplus was much less than we might think initially. And yet, even under these conditions, farmers were able to provide surplus for community storage, as well, which eventually led to centralized administration of those shared resources, and complex bureaucracies responsible for collecting, storing, and distributing grain.

## **Risk Management**

Storage, however, is not the only solution to the risk of famine in a future year. If one risk is drought, then farmers can plant not only crops that require typical rainfall to grow and produce fruit, but also drought-resistant crops that thrive even under very dry conditions. If the threat is a disaster—such as flood or hail—that can destroy a crop in the field, one solution is to plant fast-growing crops that can grow in the off-season, giving a farmer multiple opportunities for harvest in a year. Most resistant to risk are farming strategies used in tropical conditions, where year-round agriculture is possible. Crops such as bananas and manioc grow year-round, making fruit or tubers available nearly continuously; the chance of starvation is low when foods are constantly available, even eliminating the need for storage. Think about this from your perspective: If the grocery store is close by and always has food, there is no need to fill up your cupboards and refrigerator. Since you can always get food quickly, your risk of starving is essentially zero.

Archaeologists study the structure of agricultural risk, including the hazards listed above (e.g., drought, flood, hail, pests), and strategies for risk management. Careful risk management might help to produce surplus or free up surplus that would otherwise need to be stored, making it available to others. By finding archaeological markers of agricultural risk management, archaeologists can identify how these often-complex economic systems worked and deduce what factors drove decision-making by farmers.